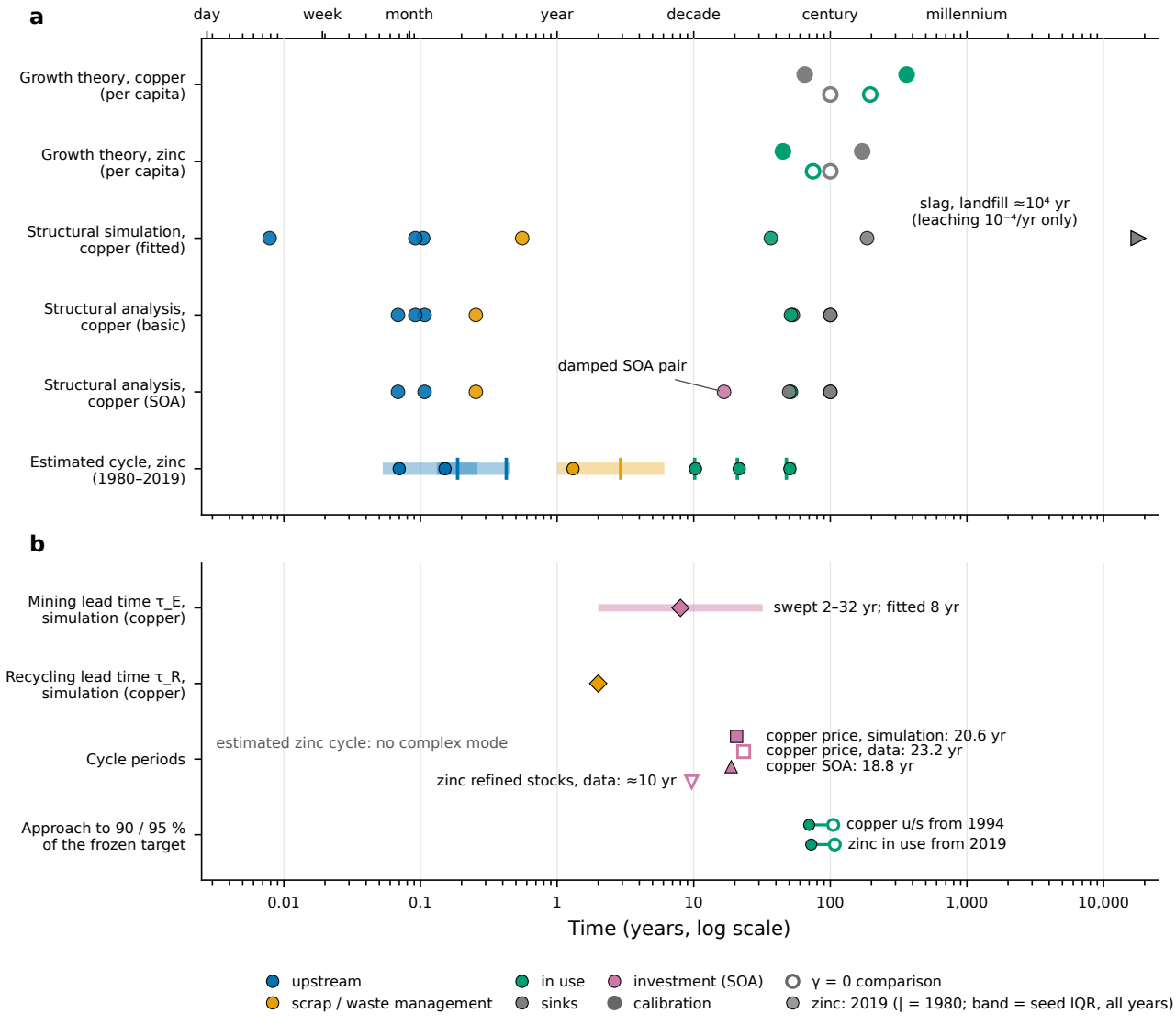
a
day
week
month
year
decade
century
millennium
Growth theory, copper (per capita)
Growth theory, zinc (per capita)
Structural simulation, copper (fitted)
Structural analysis, copper (basic)
Structural analysis, copper (SOA)
Estimated cycle, zinc (1980–2019)
slag, landfill ≈10⁴ yr (leaching 10⁻⁴/yr only)
damped SOA pair
b
Mining lead time τ_E, simulation (copper)
Recycling lead time τ_R, simulation (copper)
Cycle periods
Approach to 90 / 95 % of the frozen target
swept 2–32 yr; fitted 8 yr
estimated zinc cycle: no complex mode
copper price, simulation: 20.6 yr
copper price, data: 23.2 yr
copper SOA: 18.8 yr
zinc refined stocks, data: ≈10 yr
copper u/s from 1994
zinc in use from 2019
0.01
0.1
1
10
100
1,000
10,000
Time (years, log scale)
upstream
scrap / waste management
in use
sinks
investment (SOA)
calibration
γ = 0 comparison
zinc: 2019 (| = 1980; band = seed IQR, all years)